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# OPERATION AND MAINTENANCE INSTRUCTIONS FOR AMARILLO "A" SERIES COOLING TOWER DRIVES

Each Amarillo Fan Drive is the result of careful design and manufacturing techniques. As with any precision machine component, proper installation, maintenance and operating procedures are imperative for long life and trouble free service. The following instructions are offered to cover most conditions. Our engineers will be pleased to assist when unusual conditions require special procedures.

**INSTALLATION:** Make sure the gear drive is filled with the proper amount and type of lubricant. Normally, gear drives are shipped with the proper amount and type of lubricant installed. Install oil if drive is shipped without oil or add oil if spillage has occurred. The initial oil level should be to the middle of the oil level sight gauge and should be maintained so that the oil level is always visible in the sight gauge window when the unit is stationary, level, and the oil is at ambient temperature.

Remove rust preventive from shaft extension and clean thoroughly. Install coupling halves. Couplings should be machined for a clearance fit. Hammering or mechanically forcing the coupling on the shaft can damage the bearings or disturb the setting of the gears and is not permissible. Interference fits are permissible if the coupling half can be heated for installation and fitted without hammering or mechanically forcing on the gear drive shaft. Check runout of aligning surfaces on both coupling halves before installing connecting members. Clean any excess anti-seize compound off of the shaft, especially near the oil seals.

Level gear drive. If shimming is required, take precautions to prevent distortion of the housing. Align driver with gear drive to obtain parallel and angular alignment. Recheck alignment after two weeks of operation.

On a "Wet Tower" installation, the breather plug located on the top of the gear housing must be removed and piped to the outside of the stack, away from the direct stream of moist air. Installation of piping to the other service openings will expedite routine maintenance and can be installed if desired. Service openings include oil fill and oil drain. Use a suitable sealer for all pipe joints and coat all exposed threads to prevent corrosion. Check all external fasteners for tightness.

**OPERATION:** No special break-in procedures are necessary except as noted under lubrication.

Each unit is subjected to a running test at the factory to assure smooth and quiet operation. Excessive noise or vibration at initial operation is an indication of one or a combination of the following: (1) Misalignment (2) Imbalance of fan or other rotating parts (3) Improperly adjusted fan blades (4) Torsional vibration (5) Unstable mounting. If noise or vibration persists, shut the unit down and correct the fault before further operation.

On installations with **two speed motors**, allow a suitable time delay before switching from high speed to low speed. The fan must be at, or below, the low speed before energizing the low speed winding.

When reversing direction of the rotation, allow the fan to come to a complete stop before restarting the motor. Standard gear drives may be operated in reverse at half speed (the reversing speed must be greater than 450 rpm) continuously without damaging the gear drive. Contact the factory for recommendations prior to running the gear drive in **reverse at full speed** for extended time.

On installations with **variable speed motors**, do not operate standard gear drives below 450 RPM motor speed. For operation at motor speeds less than 450 RPM, single reduction fan drives require Amarillo Gear's exclusive **"Low Speed"** option, and double reduction fan drives require an **electric** oil pump for proper lubrication. The addition of these options allows operation with no minimum speed restrictions.

On most fan drive systems, it is not uncommon for one or more resonant speeds to exist between 0 and 1800 RPM. Continued operation at a resonant speed condition will result in torsional vibration which can be damaging to all components of the system. The most common indicator of torsional vibration is an unusual rumbling or grinding noise from the gear drive at a sharply defined speed. The noise will disappear when the speed is increased or decreased. The noise is not indicative of a defect but results when the vibratory torque exceeds the drive torque causing the gear teeth to separate and clash together very rapidly. On **variable speed** applications, operation within ±10% of a resonant speed should be avoided and the transition through a resonant speed range should be swift.

**MAINTENANCE:** Check oil level weekly with each unit stopped. Add oil if level is below oil level indicator on the double reduction, and if it is below the oil fill outlet on the single reduction.

Daily visual inspections and observation for oil leaks and unusual noises and vibrations are recommended. If any of these occurs, the unit should be shut down and the cause found and corrected.

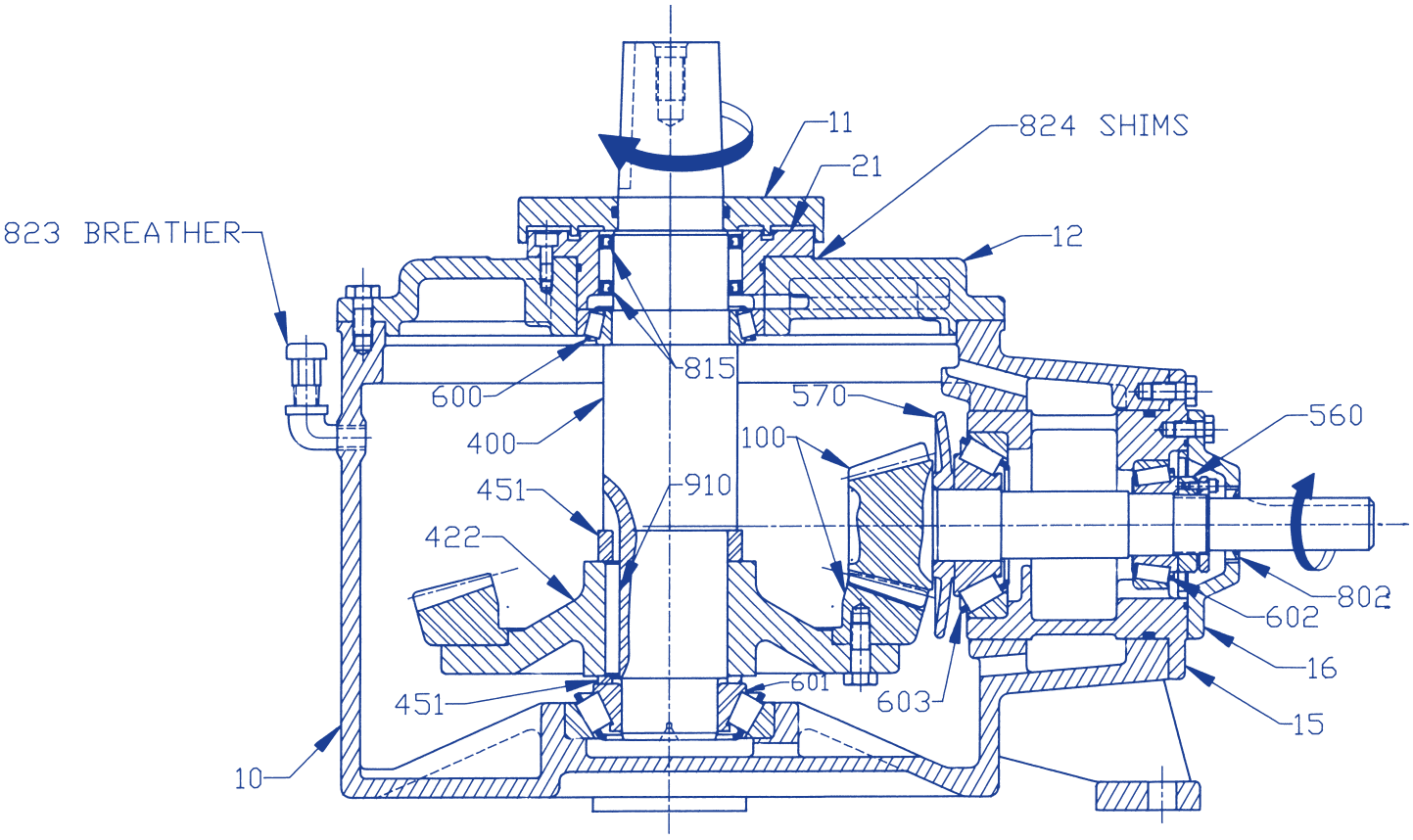
Periodic checks should be made of the alignments of all components of the system. Also, all external fasteners should be checked for tightness.

Special precautions are necessary during periods of inactivity in excess of one week. When the internal parts are not continually bathed by the lubricant as during operation, the gear drive is particularly vulnerable to attacks by rust and corrosion. For best results, let the drive cool for approximately four hours after shutdown, start the fan and let run for approximately five minutes. This will coat the internal parts of the drive with cool oil. Thereafter, run the fan for five minutes once a week throughout the shutdown period to maintain the oil film on the internal parts of the gear drive.

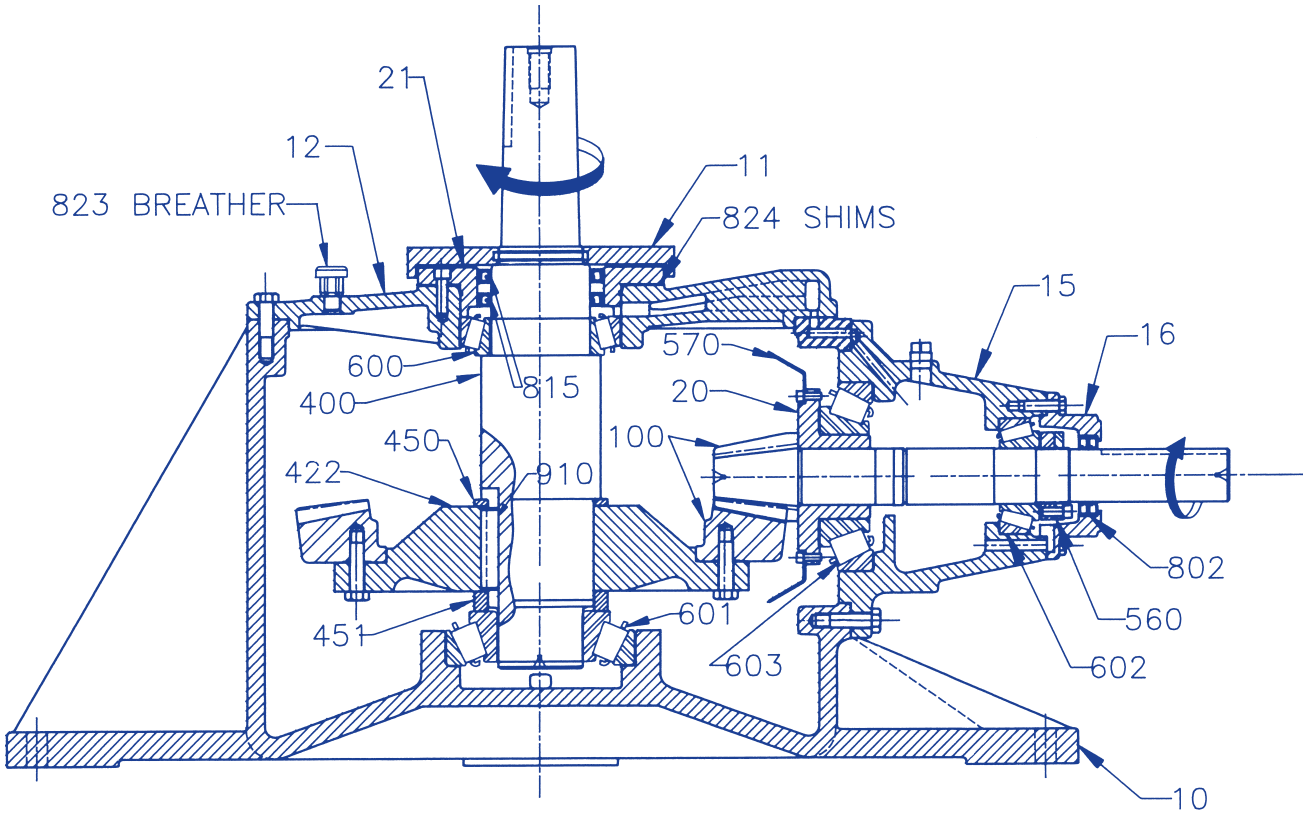
If the drive is to be inactive for a prolonged period, it is recommended that the unit be completely filled with oil. This can be accomplished by filling through the air breather port. Cover the drive with a tarpaulin or other protective covering. Drain excess oil before returning the gear drive to service.

By following the above procedures, each Amarillo Fan Drive will provide years of useful service. In the event repairs are necessary, most parts are stocked at the factory and normally can be shipped within twenty-four hours after receipt of an order. Prompt factory re-build service is also available.

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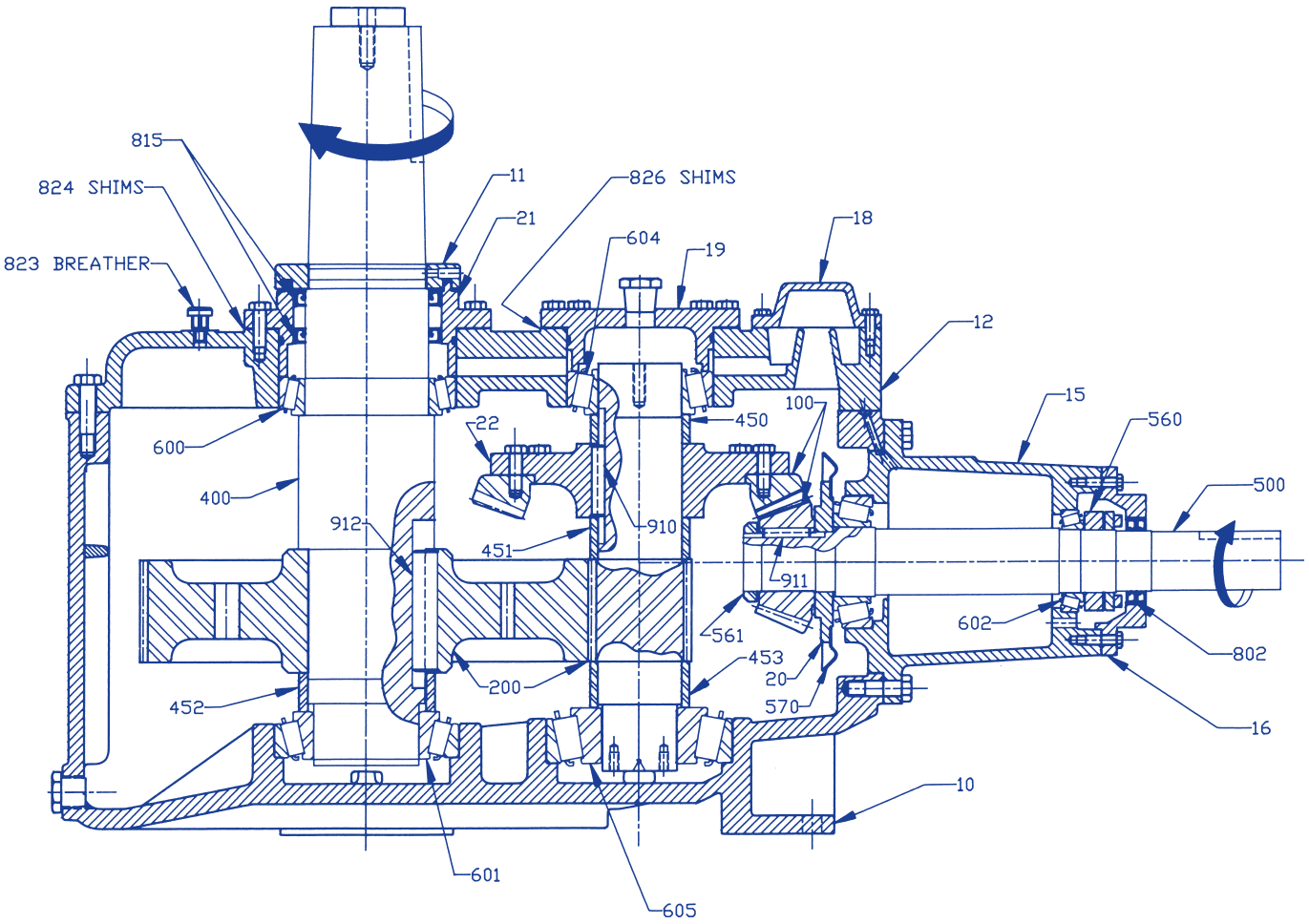



MODEL A22




MODEL A27

# DOUBLE REDUCTION



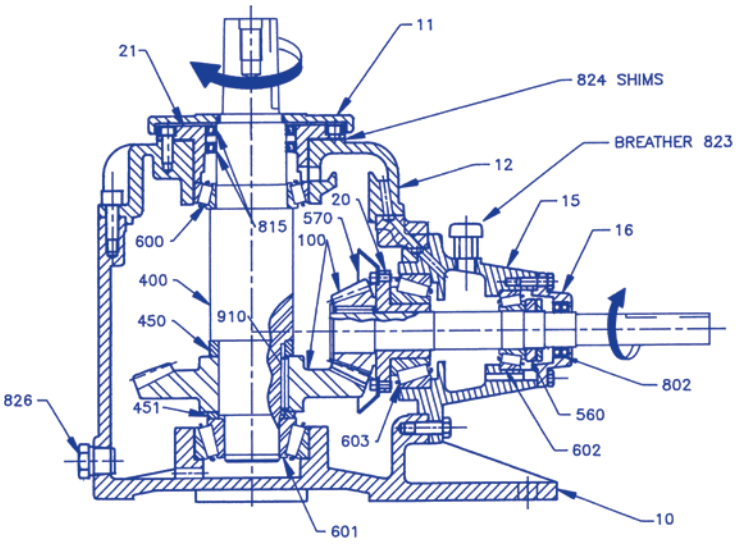

## MODEL A32, A34, A36, A38 & A40

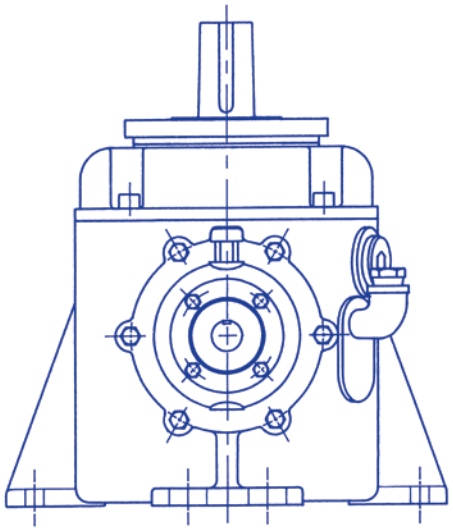
| REF NO. | PART NAME | PART NUMBER | | | | |
|---|---|---|---|---|---|---|
| | | MODEL A32 | MODEL A34 | MODEL A36 | MODEL A38 | MODEL A40 |
| 10 | (5) Gear Case | MD3–10 | MD4–10 | MD4–10 | MD4–10 | MD4–10 |
| 11 | Upper Seal Plate | MD3–11 | D4–11 | 34084 | 34084 | 34084 |
| 12 | (5) Cover | MD3–12 | MD4–12 | MD4–12 | MD4–12 | MD4–12 |
| 15 | Horizontal Carrier | MD3–15 | D4–13 | D4–13 | D4–13 | D4–13 |
| 16 | Horizontal Seal Plate | D3–16A | D5–16A | D5–16A | D5–16A | D5–16A |
| 18 | Oil Pocket Cover Plate | D2–18 | D4–18 | D4–18 | D4–18 | D4–18 |
| 19 | Upper Brg. Cover (Interm. Shaft) | D3–19 | D4–19 | D4–19M | D4–19M | D4–19M |
| 20 | Oil Slinger Plate | 33018 | D4–20 | D4–20 | D4–20 | D4–20 |
| 21 | Vertical Seal Carrier | D3–21 | D4–21 | 34085 | 34085 | 34085 |
| 22 | (3) Gear Mount | D3–22 | D4–22 | D4–22 | D4–22 | D4–22 |
| –– | Inspection Plate | L18B | L18B | L18B | L18B | L18B |
| 100 | *Spiral Bevel Gear Set | See Note (2) | See Note (2) | See Note (2) | See Note (2) | See Note (2) |
| 200 | *Helical Gear Set | 200–A32 | 200–A34 | 200–A36 | 200–A38 | 200–A40 |
| 400 | Output Shaft | See Note (6) | See Note (6) | See Note (6) | See Note (6) | 38019 |
| 450 | Above Gear Spacer (Interm. Shaft) | 450–A32 | 450–A34 | 450–A36 | 450–A38 | 450–A40 |
| 451 | Between Gear Spacer (Inerm. Shaft) | 451–A32 | 451–A34 | 451–A36 | 451–A38 | 451–A40 |
| 452 | Below Gear Spacer (Output Shaft) | 452–A32 | 452–A34 | 452–A36 | 452–A38 | 452–A40 |
| 453 | Below Gear Spacer ( Interm. Shaft) | 453–A32 | 453–A34 | 453–A36 | 453–A38 | 453–A40 |
| –– | *Between Helical Spacer | N/A | N/A | XF–A36 | XF–A38 | XF–A40 |
| 500 | (4) Horizontal Shaft | 500–A32 | 500–A34 | 500–A36 | 500–A38 | 500–A40 |
| 560 | Horizontal Locknut Set | 5237 | 6217 | 6217 | 6217 | 6217 |
| 570 | Oil Slinger | 32025 | 34022 | 34022 | 34022 | 34022 |
| 600 | *Upper Brg. (Output Shaft) | 600–A32 | 600–A34 | 600–A36 | 600–A38 | 600–A40 |
| 601 | *Lower Brg. (Output Shaft) | 601–A32 | 601–A34 | 601–A36 | 601–A38 | 601–A40 |
| 602 | *Outer Brg. (Horiz. Shaft) | 602–A32 | 602–A34 | 602–A36 | 602–A38 | 602–A40 |
| 603 | *Inner Brg. (Horiz. Shaft) | 603–A32 | 603–A34 | 603–A36 | 603–A38 | 603–A40 |
| 604 | *Upper Brg. (Interm. Shaft) | 604–A32 | 604–A34 | 604–A36 | 604–A38 | 604–A40 |
| 605 | *Lower Brg. (Interm. Shaft) | 605–A32 | 605–A34 | 605–A36 | 605–A38 | 605–A40 |
| 802 | *Horizontal Seal Set | 802–A32 | 802–A34 | 802–A36 | 802–A38 | 802–A40 |
| 815 | *Vertical Seal Set | 815–A32 | 815–A34 | 815–A36 | 815–A38 | 515–A40 |
| 823 | Breather Plug | 823 | 823 | 823 | 823 | 823 |
| 824 | *Brg. Shims (Output Shaft) | 824–A32 | 824–A34 | 824–A36 | 824–A38 | 824–A40 |
| 825 | *Gasket Set | 825–A32 | 825–A34 | 825–A36 | 825–A38 | N/A |
| 826 | *Brg. Shims (Interm. Shaft) | 826–A32 | 826–A34 | 826–A36 | 826–A38 | 826–A40 |
| –– | *Upper Brg. Cover O–Ring | M AS255 | M AS258V | M AS258V | M AS258V | M AS258V |
| –– | *Vertical Seal Carrier O–Ring | M AS364 | M AS369 | M AS267V | M AS267V | M AS267V |
| 910 | Gear Key (Interm. Shaft) | 910–A32 | 910–A34 | 910–A36 | 910–A38 | 910–A40 |
| 911 | Gear Key (Horizontal Shaft) | 911–A32 | 911–A34 | 911–A36 | 911–A38 | 911–A40 |
| 912 | Gear Key (Output Shaft) | 912–A32 | 912–A34 | 912–A36 | 912–A38 | 912–A40 |
| 913 | (4) Oil Slinger Key) | 913–A32 | 913–A34 | 913–A36 | 913–A38 | 913–A40 |
| –– | Oil Sight Gauge | P1022–6 | P1022–6 | P1022–6 | P1022–6 | P1022–6 |
| –– | Hub Retainer Plate | 37009 | 37009 | 34088 | 34088 | 34088 |
| –– | Brg. Adjustment Plate | N/A | N/A | 34086 | 34086 | 34086 |
| –– | Snap Ring | N/A | N/A | 3100–650 | 3100–650 | 3100–650 |

*RECOMMENDED SPARE PARTS

1) Specify Serial Number when ordering repair parts.
2) Spiral Bevel Gear Sets are specified by Model Number and Ratio.
3) Vertical Gear Mounts are not used for all ratios. The Gear Mount and corresponding Vertical Shaft should be ordered assembled as a set.
4) Not applicable to all ratios; consult factory
5) Gear Case and Cover must be purchased as a unit.
6) Contact factory for specific part number.

# SINGLE REDUCTION



## MODEL A20

| REF NO. | PART NAME | PART NUMBER | | |
|---|---|---|---|---|
| | | MODEL A20 | MODEL A22 | MODEL A27 |
| 10 | Gear Case | MF2–10 | MF3–10 | MF4–10 |
| 11 | Upper Seal Plate | MF2–11 | MF3–11 | MF4–11 |
| 12 | Vertical Carrier | F2–12 | MF3–12 | MF4–12 |
| 15 | Horizontal Carrier | F2–15 | MF3–15 | F5–15 |
| 16 | Horizontal Seal Plate | F2–16A | MF3–16 | F6–16A |
| 18 | Inspection Plate | N/A | N/A | E18 |
| 20 | Bearing Bushing | F2–20 | N/A | F5–20 |
| 21 | Vert. Seal Carrier | MF2–21 | MF3–21 | MF4–21 |
| 400 | Vertical Shaft | See Note (4) | See Note (4) | See Note (4) |
| 422 | Gear Mount (See Note 3) | N/A | F4–22 | L22 |
| 450 | Above Gear Spacer | 450–A20 | 450–A22 | 450–A27 |
| 451 | Under Gear Spacer | 451–A20 | 451–A22 | 451–A27 |
| 560 | Horiz. Locknut Set | 1231 | 2230 | 3268 |
| 570 | Oil Slinger | 22008 | MF3–20 | 24008 |
| 600 | *Vertical Upper Bearing | 600–A20 | 600–A22 | 600–A27 |
| 601 | *Vertical Lower Bearing | 601–A20 | 601–A22 | 601–A27 |
| 602 | *Horiz. Outer Bearing | 602–A20 | 602–A22 | 602–A27 |
| 603 | *Horiz. Inner Bearing | 603–A20 | 603–A22 | 603–A27 |
| 802 | *Horizontal Seal Set | 802–A20 | 802–A22 | 802–A27 |
| 815 | *Vertical Seal Set | 815–A20 | 815–A22 | 815–A27 |
| 823 | Breather Plug | 823 | 823 | 823 |
| 824 | *Bearing Shim Set | 824–A20 | 824–A22 | 824–A27 |
| 825 | *Gasket Set(Not Shown) | 825–A20 | N/A | 825–A27 |
| 826 | Oil Drain Plug | 826–A20 | 826–A22 | 826–A27 |
| 910 | Gear Key | 910–A20 | 910–A22 | 910–A27 |
| 100 | *Spiral Bevel Gear Set | See Note (2) | See Note (2) | See Note (2) |
| –– | *Horizontal Shim Set | N/A | SMH–A22 | N/A |
| –– | *Upper Seal Plate O–Ring | M AS140 | M AS334 | M AS337 |
| –– | *Vert. Seal Carrier O–Ring | N/A | M AS247 | M AS161 |
| –– | *Oil Seal Plate O–Ring | N/A | M AS155 | N/A |
| –– | *Horiz. Carrier O–Ring | N/A | M AS358 | N/A |

1) Specify Serial Number when ordering repair parts.
2) Spiral Bevel Gear Sets are specified by Model Number and Ratio.
3) Vertical Gear Mounts are not used for all ratios. Gear Mount and corresponding Vertical Shaft should be ordered assembled as a set.
4) Contact factory for specific part number.

*RECOMMENDED SPARE PARTS

***RECOMMENDED MINERAL OILS**

| AMBIENT TEMPERATURE AT GEAR DRIVE | 20˚ F to 120˚ F (-7˚ C to 49˚ C) |
|---|---|
| AGMA LUBRICANT NUMBER | 5 |
| ISO Grade | 220 |
| Atlantic Richfield Co. | Duro 220 |
| Chevron/Texaco | Rando HD 220, Regal R & O 220 |
| Cities Service Oil Co. | Citigo Pacemaker 220 |
| Conoco/Phillips 66/76 | Multipurpose R & O Oil 220 |
| Exxon Company | Teresstic 220 |
| Gulf Oil Corp. | Harmony 220 |
| Mobil Oil Corp. | DTE Oil BB |
| Pennzoil | Pennzbell TO 220 |
| Shell Oil Co. | Morlina S3 BA 220 |
| Sun Oil Co. | Sunvis 9220 |

*LIST OF BRAND NAMES IS FOR PURPOSE OF IDENTIFYING TYPES AND IS NOT TO BE CONSTRUED AS EXCLUSIVE RECOMMENDATIONS.

***RECOMMENDED SYNTHETIC LUBRICANTS**

| AMBIENT TEMPERATURE AT GEAR DRIVE | -20˚ F to 150˚ F (-29˚ C to 66˚ C) |
|---|---|
| AGMA LUBRICANT NUMBER | 5S |
| ISO GRADE | 220 |
| Chevron/Texaco. | Clarity 220 Synthetic |
| Conoco/Phillips 66/76 | Syncon 220 – R & O Oil |
| Mobil | SHC 630 or SHC 630* |
| Shell | Morlina S4 B 220 |

**LUBRICATION:** Use only Rust and Oxidation inhibited Gear Oils in accordance with AGMA (American Gear Manufacturers Association). Standard 9005-EO2 (or most recent edition of the standard.) For general operating conditions, use a lubricant having an AGMA lubricant number of 5. Gear oils containing Extreme Pressure (EP) additives are not recommended, and should never be used on gear drives equipped with the nonreverse option.

If the gear drive is started when the ambient temperature is below 20˚ F (-7˚ C), use a lube oil heater or a recommended synthetic oil. Lube oil heaters and synthetic oil are extra cost accessories that can be ordered with new gear drives or installed in the field.

**SYNTHETIC LUBRICANTS:** Synthetic lubricants offer advantages of extended service life, a broader operational temperature range, reduced friction, and the ability to maintain a higher film strength which can extend the service life of the gear drive. When the operating temperature exceeds 180˚ F (82˚ C) or the gear drive is started when the ambient temperature is below 20˚ F (-7˚ C), a synthetic lubricant is recommended. Synthetic lubricants can be made of various base stocks which are incompatible with certain gear drive components; therefore, any synthetic lubricant not listed in this bulletin should be approved by Amarillo Gear Company. Do not use synthetic lubricants made from ester base stocks. Change intervals for synthetic lubricants should not be extended beyond the change interval for mineral oils without a comprehensive monitoring program.

**CHANGE INTERVAL:** The original oil should be replaced after 500 hours of operation or four weeks, whichever comes first. It is recommended that the oil be drained when it is at or near operating temperature. Refill the drive with the recommended type and amount of lubricant.

Normally the oil should be changed every 2500 hours or every six months, whichever comes first. Shorter change intervals of two to three months may be required if the gear drive is subjected to unusual operating conditions such as very moist atmosphere, rapid temperature change, consistent high operating temperature or any conditions that tend to contaminate the oil or promote the formulation of sludge and deposits inside the gear case.

The vertical and horizontal shafts are equipped with grease lubricated dual seals. Relubrication is not required.

**OIL CAPACITY**

SINGLE REDUCTION DRIVES

| MODEL | GALLONS | LITERS |
|---|---|---|
| A20 | 2 | 7.5 |
| A22 | 4 | 15 |
| A27 | 10 | 38 |
| | | |

DOUBLE REDUCTION DRIVES

| MODEL | GALLONS | LITERS |
|---|---|---|
| A32 | 17 | 64 |
| A34 | 21 | 79 |
| A36 | 21 | 79 |
| A38 | 21 | 79 |
| A40 | 21 | 79 |